

Master Thesis
BlueRov2 Formation Control

Week 3

João Zenário

Plan For Third Week

- Formation control with single integrator agents no constraints
  - Undirected graph
  - Directed graph
- Single integrator formation control with constraints
  - Only connectivity maintenance
  - Then also colision avoidance
- Double integrator agents

- BlueRov2 model no input constraints
  -
- [Research challenge] integrate input constraints
  -
- [Research challenge] controller robust or less sensitive to uncertainties (e.g. inverse optimal controller in the sense of LgV like control laws rather than feedback linearization)

Achievements

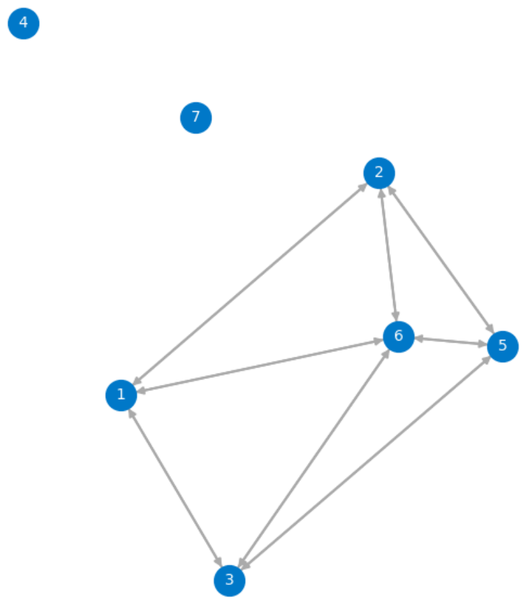
- Study
  - Graph Theoretic Methods in Multiagent Networks
  - Esteban’s thesis
  - Khalil's book on Nonlinear Control

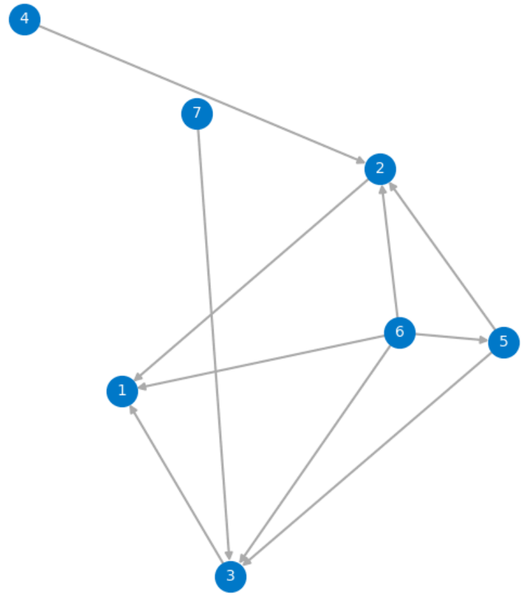
- Implementation
  - Improvements
    - Implemented system stability (ensured spanning tree)
    - Added distance and velocity graphs

Improvements - System Stability

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$





Improvements - System Stability

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

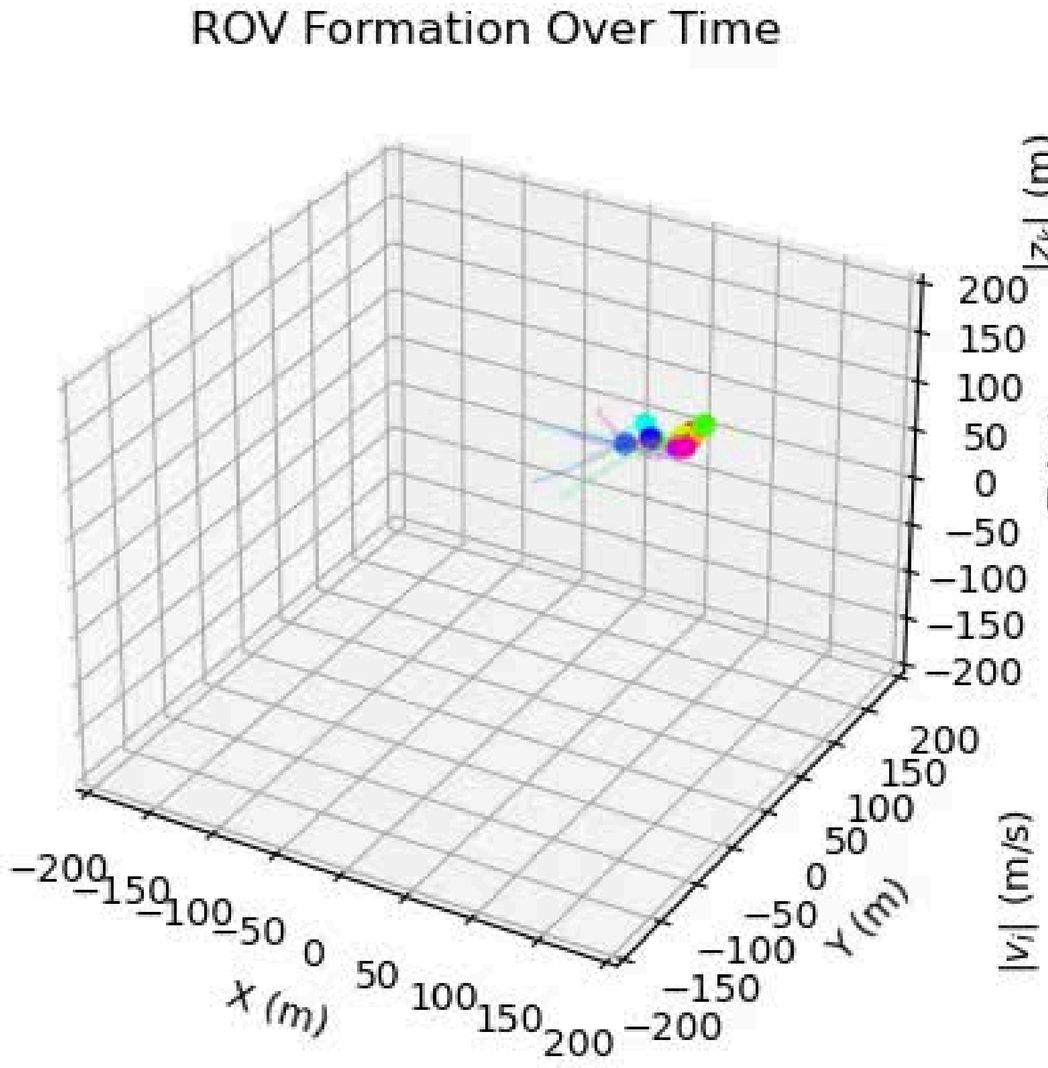
$$D = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

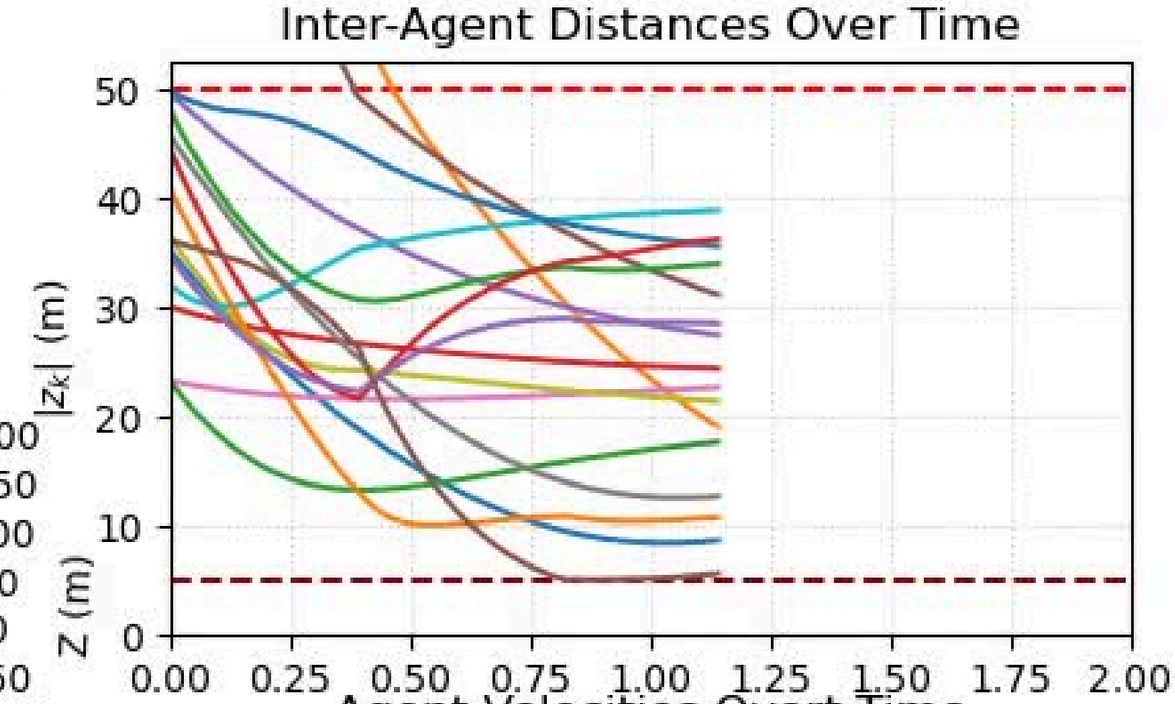
For a directed graph, all non-zero eigenvalues of L have positive real parts and 0 is a simple eigenvalue if and only if the directed graph contains a directed spanning tree

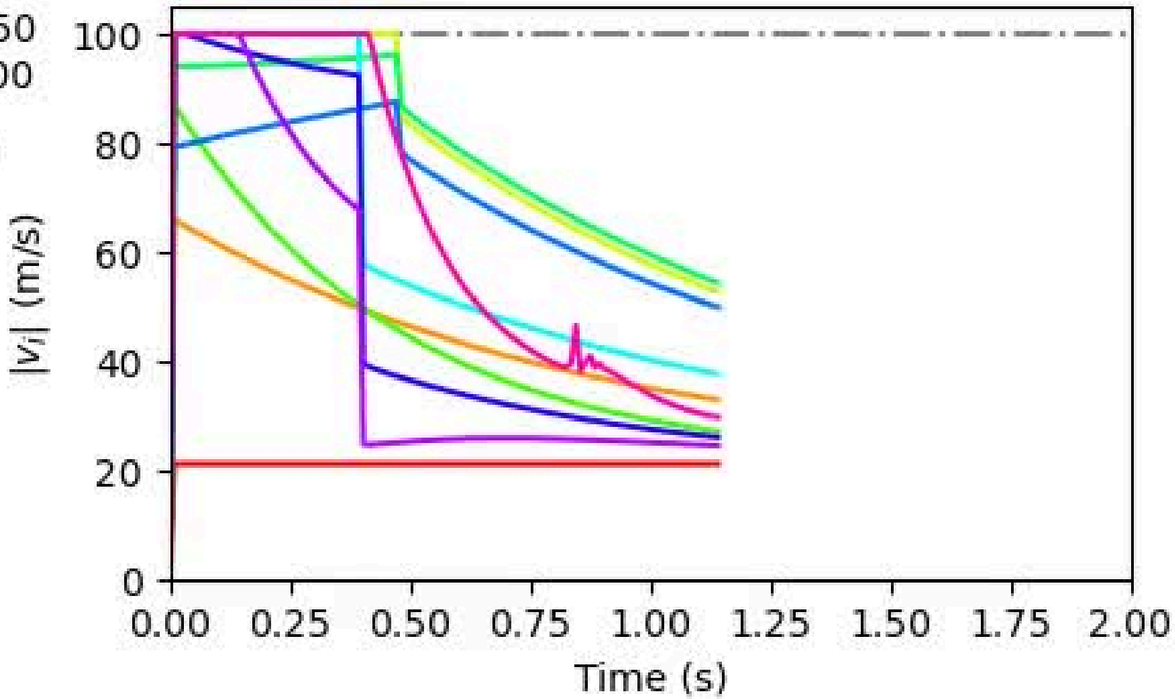
$$L = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 2 & 0 & 0 \\ -1 & -1 & -1 & 0 & -1 & 4 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\lambda(L) = \{0, 1, 1, 1, 1, 2, 4\}$$

Improvements - Added Graphs







Next Steps

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Questions & Comments